Adobe

PRD: Adobe Competitive Intelligence Automation

Status: Active — Phase 1 Complete

Version: 2.0

Last Updated: May 2026

Repo: github.com/bnamatherdhala7/Competitive-research-Report-using-Claude

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1. SUMMARY

An end-to-end system that generates branded, acquisition-focused competitive intelligence PDFs for Adobe product teams (Acrobat, Express, Firefly). Phase 1 is live: three full playbooks covering pricing, messaging, battlecards, and PLG motions are generated by Claude using live web research and exported as Adobe-branded PDFs in under 60 seconds. Phase 2 delivers reports automatically to Slack channels on a monthly schedule. Phase 3 builds a full intelligence platform with trend tracking, dashboard, and cross-team distribution.

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2. CONTACTS

Name	Role	Responsibility
Bharat Namatherdhala	**Adobe PM (Owner)**	**Requirements, report consumer, stakeholder demos**
Claude Sonnet 4.6	AI Agent (Builder)	Code, automation, ongoing maintenance

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3. HOW A PDF REPORT IS GENERATED — FULL PIPELINE

There are two modes: **Automated** (runs in the cloud via Trigger.dev) and **Manual** (runs locally via Claude Code). Both produce the same output.

Mode A — Manual Pipeline (Claude Code, used for all current playbooks)

```

PM types: "generate acquisition playbook for Adobe Acrobat"

↓

Claude Code reads CLAUDE.md + workflow SOP

(decides what sections to write, what data to find)

↓
```

API Call 1: WebSearch (Brave Search API)

→ Query: "Adobe Acrobat pricing 2026"

→ Query: "Foxit PDF Editor pricing 2026"

→ Query: "UPDF vs Adobe Acrobat comparison"

→ Query: "Adobe Acrobat churn complaints Reddit"

→ Returns: up to 10 live web results per query

↓

Claude Sonnet 4.6 synthesizes results into 13-section markdown

(pricing tables, battlecards, PLG motions, metrics dashboard)

↓

Claude writes: docs/Acrobat-Acquisition-Playbook.md

↓

Script call: npx tsx scripts/md-to-pdf.ts docs/Acrobat-Acquisition-Playbook.md

↓

md-to-pdf.ts (Puppeteer):

→ Reads markdown file

→ Converts markdown → HTML with inline CSS (system fonts, Adobe red header)

→ Puppeteer launches headless Chrome

→ page.setContent(html, { waitUntil: "networkidle0" })

→ page.pdf({ format: "A4", printBackground: true })

→ Saves PDF → ~/Desktop/Adobe-Acrobat-Acquisition-Playbook-2026-04.pdf

↓

PM opens PDF on Desktop

```

**APIs called per report (manual mode):**

| API | Purpose | Cost | Auth |
|---|---|---|---|
| **Brave Search API** | **Live web research (pricing, competitor sites, Reddit threads)** | **Free tier (2,000 queries/month)** | **`BRAVE_API_KEY` in `.env`** |
| Anthropic Claude Sonnet 4.6 | Synthesis, writing, structuring the full report | ~$0.08–0.15/report | `ANTHROPIC_API_KEY` in `.env` |
| Puppeteer (headless Chrome) | HTML → PDF rendering | Free (local) | None |

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### Mode B — Automated Pipeline (Trigger.dev, used for scheduled Firefly reports)

```

Trigger.dev CRON fires: 1st of every month, 9am PT

↓

orchestrator.ts (main task) starts

```
↓
┌─ batchTriggerAndWait([
| reddit.ts → Reddit public JSON API (no key needed)
| searches: "{competitor} credits pricing"
| returns: top 3 posts per competitor
|
| youtube.ts → YouTube Data API v3
| searches: "{competitor} AI image review 2026"
| returns: 3 recent video titles + descriptions
|
| web-search.ts → Brave Search API
| searches: "{competitor} pricing site:{competitor}.com"
| returns: 5 results per competitor
| ])
└─ (all 7 competitors scraped in parallel, ~15 seconds)
↓
analyzer.ts → Anthropic Claude Haiku
→ Input: all scraped data, hard-capped at 6,000 tokens
→ Prompt: structured JSON schema for competitive analysis
→ Output: JSON with executive summary, pricing matrix, recommendations
↓
report-template.ts → builds HTML from JSON
→ Adobe brand colors, dark canvas theme, structured sections
↓
Trigger.dev stores run output (accessible via Management API)
↓
(Manual step) PM runs: npm run save-report
→ save-report.ts fetches latest run output via Trigger.dev API
→ Puppeteer renders HTML → PDF
→ Saves to ~/Desktop/
```

APIs called per report (automated mode):

API	Purpose	Cost	Auth
Reddit public JSON API	**Community sentiment on competitor credit packs**	**Free (no key)**	**None**
YouTube Data API v3	Video reviews and tutorials per competitor	Free (10K units/day)	`YOUTUBE_API_KEY`
Brave Search API	Live pricing and news per competitor	Free (2K queries/month)	`BRAVE_API_KEY`
Anthropic Claude Haiku	Analysis synthesis — cheapest capable model	~$0.003/report	`ANTHROPIC_API_KEY`
Trigger.dev Management API	Fetch run output for PDF generation	Included in plan	`TRIGGER_API_KEY`
Puppeteer (headless Chrome)	HTML → PDF	Free (local)	None

Total automated cost per report: ~$0.003 — less than one-third of one cent.

PDF Generation Deep Dive (scripts/md-to-pdf.ts)

The PDF script does the following in order:

```

1. Reads markdown file from disk (readFileSync)
2. Parses markdown → HTML:

- Headings (#, ##, ###) →

'

---

**- Bold/italic → ,**

***- Tables (|col|col|) →***

***- Lists (-, 1.) →***

,

***- Blockquotes (>) →***
```

(used for TL;DR callout box)

1. Wraps in full HTML document with inline CSS:

- Font: system fonts only (-apple-system, BlinkMacSystemFont, Segoe UI, Arial)

→ No Google Fonts = no external requests = PDF renders reliably

- Theme: white background, #1E1E1E text

- Adobe header: red (#EB1000) background, "Adobe" logo, report title

- Tables: dark header row, alternating row shading

- Print CSS: @page { size: A4; margin: 0 }

1. Puppeteer launches headless Chrome

2. page.setContent(html, { waitUntil: "networkidle0" })

→ waitUntil: "networkidle0" ensures all rendering is complete before capture

1. page.pdf({ format: "A4", printBackground: true, margin: 0 })

2. Saves to ~/Desktop/Adobe-{filename}-{YYYY-MM}.pdf

```

*Why system fonts? Earlier versions loaded Google Fonts ( `Source Sans 3` ) via a  tag. Headless Chrome cannot reliably load external fonts before PDF capture, causing blank/empty PDFs. System fonts render instantly with no network dependency.*

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## 4. MCP INTEGRATIONS

*MCP (Model Context Protocol) allows Claude to call external tools and services as part of its reasoning loop — without writing separate glue code. The following MCPs are used or planned:*

*Currently Active*

| MCP | What it does in this system | Where used |
|---|---|---|
| *WebSearch (Brave)* | *Claude calls live web search mid-conversation to verify current pricing, find competitor pages, and pull real data before writing reports* | *All manual playbook generation* |
| *Filesystem* | *Claude reads/writes files directly — markdown reports, markdown sources, config files* | *Writing `.md` files, reading workflow SOPs* |
| *Bash/Shell* | *Claude runs `npx tsx scripts/md-to-pdf.ts` directly from the conversation to trigger PDF generation* | *PDF generation step* |
```

Phase 2 — Planned MCP Integrations

MCP	Purpose	Required for
Slack MCP	*Post formatted report summaries to `#competitive-intel` channel with PDF attachment*	*Phase 2 Slack delivery*
Google Drive MCP	*Save PDFs to a shared Drive folder; auto-generate shareable link*	*Phase 2 distribution*
GitHub MCP	*Commit new report files, create PRs for report updates without leaving the Claude session*	*Phase 2 CI/CD*
Email (Resend MCP)	*Send report to distribution list on the 2nd of each month*	*Phase 2 distribution*

Phase 3 — Advanced MCP Integrations

MCP	Purpose
Notion MCP	*Sync report sections directly into a Notion competitive intel database*
Linear MCP	*Auto-create PM tickets from PLG recommendations in each report*
Airtable MCP	*Log per-competitor metrics to a shared Airtable base for trend tracking*
Confluence MCP	*Publish reports to Adobe's internal wiki for broader team access*

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5. CURRENT DELIVERABLES (PHASE 1 COMPLETE)

All three PDF playbooks are live in the repo and downloadable at the Vercel site.

Report	File	Pages	Key Sections
Acrobat Acquisition Playbook	*`Adobe-Acrobat-Acquisition-Playbook-2026-04.pdf`*	*~8*	*Pricing · 3 Battlecards · PLG Now/3M/6M · 8 Metrics*
Express Acquisition Playbook	*`Adobe-Express-Acquisition-Playbook-2026-04.pdf`*	*~8*	*Pricing · vs Canva/M365/Picsart · PLG loops · 8 Metrics*
Remove Background Playbook	*`Adobe-Remove-Background-Playbook-2026-04.pdf`*	*~8*	*18M keyword · SEO sprint · Shopify app · 8 Metrics*
Firefly Automated Report	*Generated monthly via Trigger.dev*	*~4*	*Competitor pricing matrix · Reddit sentiment · PM recommendations*

Markdown sources (in `docs/`):

- *`docs/Acrobat-Acquisition-Playbook.md`*
- *`docs/Express-Acquisition-Playbook.md`*
- *`docs/Remove-Background-Playbook.md`*

Demo resources (in `docs/`):

- *`docs/DEMO-GUIDE.md` — 10-minute demo script + Q&A prep + "how to use in your Claude account"*

6. PHASE ROADMAP

Phase 1 — Local Generation (COMPLETE ✅)

Goal: Replace manual competitive research with AI-generated PDFs on demand.

What was built:

- *Reddit + Brave + YouTube scraper for Firefly competitors (automated via Trigger.dev)*
- *Claude-generated acquisition playbooks for Acrobat, Express, and Remove Background keyword*
- *`md-to-pdf.ts` — markdown → Adobe-branded PDF in one command*
- *CLAUDE.md + workflow SOPs — plain-English instructions that make Claude a consistent agent*
- *GitHub repo + Vercel deployment — PDFs downloadable at public URL*
- *DEMO-GUIDE.md — 10-minute demo for org presentation*

How to run:

```bash
npm run test-run

npx tsx scripts/md-to-pdf.ts docs/[filename].md
```

Cost: ~$0.003–0.15/report. Under $2/year at monthly cadence.

Phase 2 — Slack Integration (NEXT)

Goal: Reports land in the right Slack channels automatically, with no manual steps after generation.

What to build:

2.1 — Automated monthly Slack delivery

Every 1st of the month, Trigger.dev fires the Firefly competitor pipeline and posts the executive summary directly to `#competitive-intel`:

```
Competitive Intel — May 2026 ⚡
```

• Midjourney raised Standard plan from $30 → $35/mo (15% increase)

• DALL-E added real-time generation; early reviews cite 3x speed vs Firefly

• Canva AI added bulk generation to Pro; community reaction mixed on quality

• Ideogram 3.0 launched with photorealism claims — Reddit positive, skeptical on commercial use

• Adobe Firefly: commercial safety remains unchallenged; CC integration praised in r/photography

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📄 *Full report: Download PDF*

🔗 *Source: github.com/bnamatherdhala7/Competitive-research-Report-using-Claude*

```

*API calls added in Phase 2:*

| New API | Purpose | Auth |
|---|---|---|
| Slack Web API (`chat.postMessage`) | Post summary to channel | `SLACK_BOT_TOKEN` in `.env` |
| Slack Files API (`files.upload`) | Upload PDF as Slack file attachment | `SLACK_BOT_TOKEN` in `.env` |
| Google Drive API (optional) | Save PDF to shared folder, generate shareable link | `GOOGLE_SERVICE_ACCOUNT_JSON` |
| Resend API (optional) | Send PDF by email to distribution list | `RESEND_API_KEY` |

*Trigger.dev task changes:*

```typescript
// src/trigger/competitor-analysis/orchestrator.ts — Phase 2 additions
// After PDF is generated:
await slack.postMessage({
channel: "#competitive-intel",
text: buildSlackSummary(report), // 5-bullet executive summary
attachments: [{ title: "Full PDF Report", ...pdfAttachment }]
});
// Optional: save to Drive
const driveLink = await uploadToGoogleDrive(pdfBuffer, reportTitle);
await slack.postMessage({ channel: "#competitive-intel", text: `📩 PDF saved: ${driveLink}` });
```

2.2 — On-demand Slack bot command

A Slack slash command `/competitive-intel [product]` triggers a new report:

```
User types: /competitive-intel Adobe Express
↓
Slack sends webhook to Trigger.dev
↓
orchestrator.ts runs Express pipeline
```
```

↓

Slack bot replies in thread: "Generating... (est. 45 seconds)"

↓

Report ready → bot posts PDF + 5-bullet summary in thread

```

*Implementation:*

- *Slack App with slash command* `/competitive-intel`
- *Slash command webhook → Trigger.dev HTTP endpoint*
- *Trigger.dev task runs pipeline → posts result back to Slack thread*

*2.3 — Targeted channel routing*

| *Report type* | *Slack channel* |
|---|---|
| *Acrobat updates* | `#acrobat-pm` |
| *Express updates* | `#express-growth` |
| *Firefly updates* | `#firefly-competitive-intel` |
| *All reports (digest)* | `#pm-competitive-intel` |

*Phase 2 effort estimate: 3–5 days of Claude Code work.*

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*Phase 3 — Intelligence Platform*

*Goal: Trend tracking, cross-team access, and a self-service dashboard — making this the source of truth for Adobe competitive intelligence across product teams.*

*What to build:*

*3.1 — Trend detection between reports*

*Each month's report is compared to the previous month. Claude highlights what changed:*

```

CHANGES DETECTED — May vs April 2026

———————————————————————

🔴 *Midjourney: price +17% (Standard $30→$35) — flagged for Firefly pricing review*

🟡 *DALL-E: launched real-time generation — competitive threat in speed positioning*

🟢 *Adobe Firefly: community sentiment improved +12% — commercial safety narrative landing*

———————————————————————

```

*Implementation:*

- *Store each report's JSON output in a database (PlanetScale or Supabase free tier)*
- *On each new run, Claude Haiku diffs current vs previous JSON*
- *Delta summary prepended to report and Slack post*
```

3.2 — Web dashboard

A simple read-only dashboard at `competitive-research-report-using-c.vercel.app/dashboard` *showing:*

- *Report archive (all past reports, downloadable)*
- *Trend charts per competitor (pricing over time, sentiment score over time)*
- *Key metrics table updated monthly*

Tech: Next.js (or plain HTML) + Vercel + Supabase for JSON storage.

3.3 — Notion / Confluence integration

Each report's key sections (TL;DR, pricing table, battlecards, PLG recommendations) are synced to a Notion database or Confluence page — so teams who live in those tools get the intelligence without needing to find the PDF.

3.4 — Linear ticket auto-creation

PLG recommendations from each report automatically create Linear issues:

```
Title: [PLG] Cancel-intervention modal — Acrobat retention

Body: From April 2026 competitive report — cancel intervention modal

can lift retention from 10% → 30%. Counter: Foxit/UPDF no-ETF pitch.

Metric: cancellation-to-retention rate.

Labels: competitive-intel, plg, acrobat
```

3.5 — Expanded product coverage

Extend the automated pipeline beyond Firefly to cover Acrobat, Express, and Remove Background keyword monthly — using the same Trigger.dev + Brave + Claude architecture, with each product's playbook template as the synthesis prompt.

Phase 3 effort estimate: 2–3 weeks of Claude Code work.

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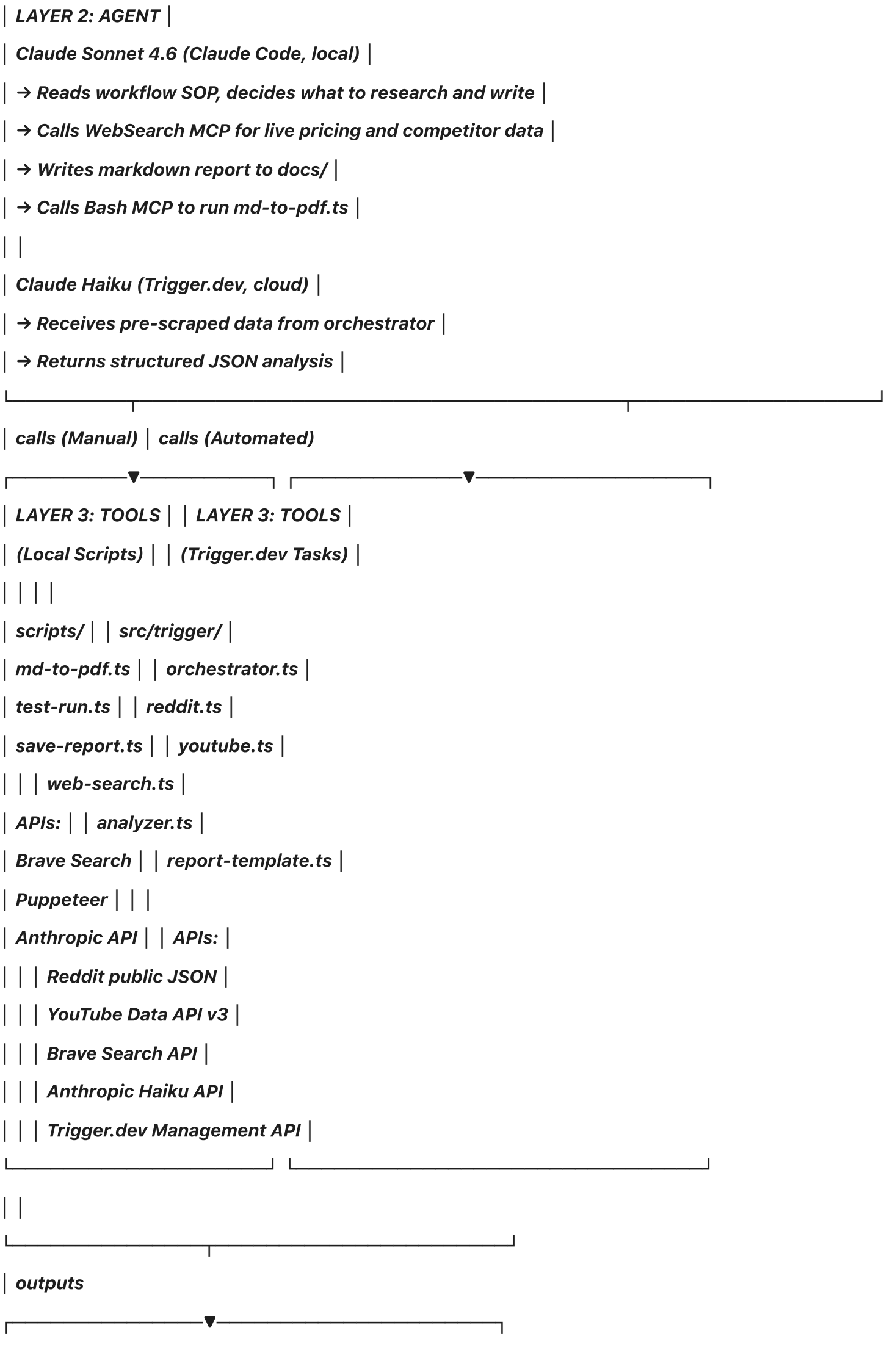
7. FULL SYSTEM ARCHITECTURE

```
┌──────────────────────────────────────────────────────────────────────┐
│ LAYER 1: WORKFLOWS                                                   │
│ workflows/competitor-analysis.md (Firefly automated SOP)             │
│ workflows/competitive-analysis-template.md (universal template)      │
│ CLAUDE.md (agent instructions, cost rules, section requirements)      │
└──────────────────────────────────────────────────────────────────────┘
                                  │
                                  │ read by
                                  ▼
```


LAYER 2: AGENT
Claude Sonnet 4.6 (Claude Code, local)
→ Reads workflow SOP, decides what to research and write
→ Calls WebSearch MCP for live pricing and competitor data
→ Writes markdown report to docs/
→ Calls Bash MCP to run md-to-pdf.ts
Claude Haiku (Trigger.dev, cloud)
→ Receives pre-scraped data from orchestrator
→ Returns structured JSON analysis
calls (Manual)
calls (Automated)
LAYER 3: TOOLS
(Local Scripts)
scripts/
md-to-pdf.ts
test-run.ts
save-report.ts
APIs:
Brave Search
Puppeteer
Anthropic API
LAYER 3: TOOLS
(Trigger.dev Tasks)
src/trigger/
orchestrator.ts
reddit.ts
youtube.ts
web-search.ts
analyzer.ts
report-template.ts
APIs:
Reddit public JSON
YouTube Data API v3
Brave Search API
Anthropic Haiku API
Trigger.dev Management API
outputs

```
│ DELIVERABLES │
│ ~/Desktop/Adobe-*.pdf (local) │
│ docs/*.pdf (repo + Vercel) │
│ docs/*.md (markdown sources) │
│ Trigger.dev run logs (cloud) │
└──────────────────────────────────────────────┘

Phase 2 adds ↓

┌──────────────────────────────────────────────┐
│ Slack: #competitive-intel channel │
│ Google Drive: shared PDF folder │
│ Email: distribution list (Resend) │
└──────────────────────────────────────────────┘

Phase 3 adds ↓

┌──────────────────────────────────────────────┐
│ Dashboard: vercel.app/dashboard │
│ Notion / Confluence sync │
│ Linear tickets from PLG recs │
│ Supabase: historical JSON storage │
└──────────────────────────────────────────────┘
```

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8. ENVIRONMENT VARIABLES

All secrets live in `.env` only — never in code, never in comments, never in logs.

Variable	Used for	Where to get it
`ANTHROPIC_API_KEY`	Claude Haiku (analysis) + Claude Sonnet (manual)	console.anthropic.com
`BRAVE_API_KEY`	Brave Search (web research)	api.search.brave.com
`YOUTUBE_API_KEY`	YouTube Data API v3 (video search)	console.cloud.google.com
`TRIGGER_API_KEY`	Trigger.dev Management API	cloud.trigger.dev
`TRIGGER_PROJECT_ID`	Trigger.dev project identifier	cloud.trigger.dev
`SLACK_BOT_TOKEN`	(Phase 2) Slack Web API	api.slack.com/apps
`SLACK_CHANNEL_ID`	(Phase 2) Target Slack channel	Slack app settings
`RESEND_API_KEY`	(Phase 2) Email delivery	resend.com
`GOOGLE_SERVICE_ACCOUNT_JSON`	(Phase 2) Google Drive upload	Google Cloud Console

9. COST SUMMARY

Mode	Per report	Monthly (1 report)	Annual (12 reports)
Manual (Claude Sonnet)	~$0.10–0.15	~$0.12	~$1.44
Automated (Claude Haiku)	~$0.003	~$0.003	~$0.036
Phase 2 (+ Slack, Drive)	+$0	Slack free tier	Included
Phase 3 (+ Supabase, Linear)	+$0	Supabase free tier	Included

Hard cap: Never exceed $0.50/report. Stop if Anthropic API costs spike.

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10. HOW TO USE THIS SYSTEM IN YOUR CLAUDE ACCOUNT

Anyone on the team with a Claude account can generate a new competitive playbook:

Option 1 — Claude.ai (no setup, 2 minutes)

Paste this into Claude at claude.ai:

> *"You are a competitive intelligence analyst for Adobe. Generate a focused acquisition playbook for [product name]. Include: TL;DR (2–3 sentences), pricing comparison table vs top 5–6 competitors with real prices, competitor website messaging, top 5 churn triggers with rank, win signals, 3 battlecards (win move + lose scenario per competitor), PLG acquisition strategy with Now/3M/6M moves (each must name the competitor countered, what to build, and what metric to track), and a PLG metrics dashboard table with current estimates and 6-month targets. Use web search to verify current pricing."*

Option 2 — Claude Code CLI (generates PDF automatically)

```bash
npm install -g @anthropic-ai/claude-code
cd path/to/Competative-Intelligence
claude
```

Option 3 — Full automated pipeline (requires API keys)

```bash
git clone https://github.com/bnamatherdhala7/Competitive-research-Report-using-Claude
cd Competitive-research-Report-using-Claude
npm install
echo "ANTHROPIC_API_KEY=your_key" >> .env
```

echo "BRAVE_API_KEY=your_key" >> .env

npm run test-run # scrape + analyze

npm run save-report # generate PDF → Desktop

```

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```